MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGULATOR, POSITIVE, POWER, KEY CODE ARM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This document establishes the requirements and test procedures for the procurement of an electronic positive voltage regulator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARD

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665		Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements All modules procured to this specification shall have a predicted failure rate no greater than 0.350 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements The module shall conform to the electrical requirement specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ARM.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20. Contact pins positioned A21 through A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A, and an additional heat dissipation fin on increment B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/87-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

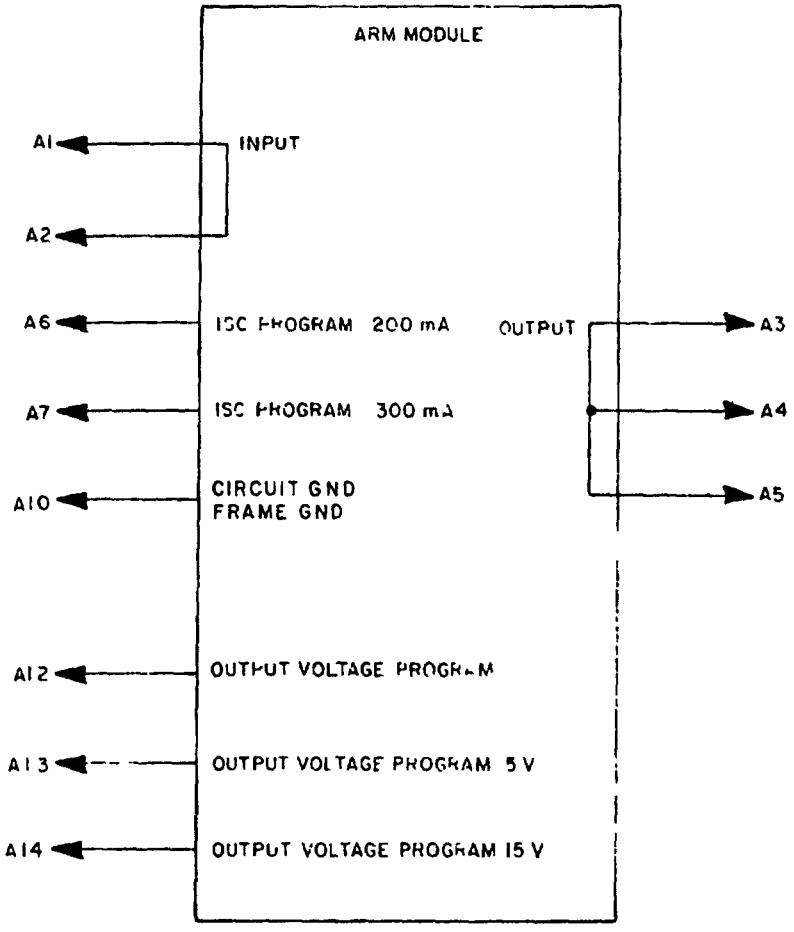
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronics Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing Test conditions A through L shall be performed in any order with the following exceptions. If test conditions A or K are part of a sequence of tests, the order of testing shall be as follows: A, K where applicable, followed by the remaining tests in the sequence

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NOTE. Unused pins: A8, A9, A11, A15, A16, A17, A18, A19, and A20.

*FIGURE 1 ARM module function diagram, positive regulator.

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TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	Isolation	II	X	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
B5	Continuity	III	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
C1	Output voltage program	IV	X	4.95	5.05	4.90	5.10	4.90	5.10	4.85	5.15	Vdc
C2	Output voltage program	IV	X	14.70	15.30	14.55	15.45	14.55	15.45	14.40	15.60	Vdc
D1	DC input voltage regulation	V	X	4.95	5.05	4.90	5.10	4.90	5.10	4.85	5.15	Vdc
D2-D7	DC input voltage regulation	V	X	← ±15 mV variance max from D1 →								
D8	DC input voltage regulation	V	X	14.70	15.30	14.55	15.45	14.55	15.45	14.40	15.60	Vdc
D9-D12	DC input voltage regulation	V	X	← ±35 mV variance max from D8 →								
E1, E2	Output short circuit current	VI	X	96.3	117.7	93.6	120.3	85.0	128.8	82.5	131.3	mAdc
E3, E4	Output short circuit current	VI	X	183.6	224.4	178.5	229.5	160.0	243.8	150.0	253.8	mAdc
E5, E6	Output short circuit current	VI	X	276.4	337.6	268.6	345.4	230.0	370.0	210.0	390.0	mAdc
E7, E8	Output short circuit current	VI	X	363.6	444.4	353.5	454.5	320.0	480.0	300.0	500.0	mAdc
F1, F2	Input modulation regulation	VII	X	0.1	750.0	0.1	750.0	0.1	750.0	0.1	750.0	μVrms
F3, F4	Input modulation regulation	VII	X	0.01	2.25	0.01	2.25	0.01	2.25	0.01	2.25	mVrms
G1	Output load regulation	VIII	X	4.95	5.05	4.90	5.10	4.90	5.10	4.85	5.15	Vdc
G2-G7	Output load regulation	VIII	X	← ±10 mV variance max from G1 →								
G8	Output load regulation	VIII	X	14.70	15.30	14.55	15.45	14.55	15.45	14.60	15.40	Vdc

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
G9-G14	Output load regulation	VIII	X	← ±30 mV variance max from G8 →								
H1, H2	Output noise	IX	X	0.0	250.0	0.0	250.0	0.0	250.0	0.0	250.0	µVrms
J1, J2	Output impedance check	X	S	5.0	500.0	5.0	500.0	5.0	500.0	5.0	500.0	µVrms
J3	Output impedance check	X	Q	10	1,000	10	1,000	10	1,000	10	1,000	µVrms
K1	Output voltage latch	XI	S	-.950	-.650	-.950	-.650	-.950	-.500	-.950	-.500	Vdc
K2	Output voltage latch	XI	S	4.95	5.05	4.90	5.10	4.90	5.10	4.85	5.15	Vdc
L1	Load transient response	XII	S	0	150	0	150	0	150	0	150	mV
L2	Load transient response	XII	S	0	450	0	450	0	450	0	450	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through J and L shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through J and L shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through L shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification Part number M28787/87-1 supersedes part number M28787/87. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	3.75 W
Input voltage	10 to 40 Vdc
Output impedance	0.100Ω maximum at 1 kHz and at 100 Hz 0.200Ω maximum at 50 kHz
Output voltage programming	5 Vdc (Jumper pin A12 to A13) 15 Vdc (Jumper pin A12 to A14)
Output current limiting programming	100 mA (No connection) 200 mA (Jumper pin A3 to A6) 300 mA (Jumper pin A3 to A7) 400 mA (Jumper pin A3 to A6 and A7)

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6.5 Module function. This module contains a positive voltage regulator programmable to two voltage levels, 5 and 15 Vdc with a maximum power dissipation of 3.75 watts at 25°C (see figure 2) The power dissipation is derated at 24 mV/°C.

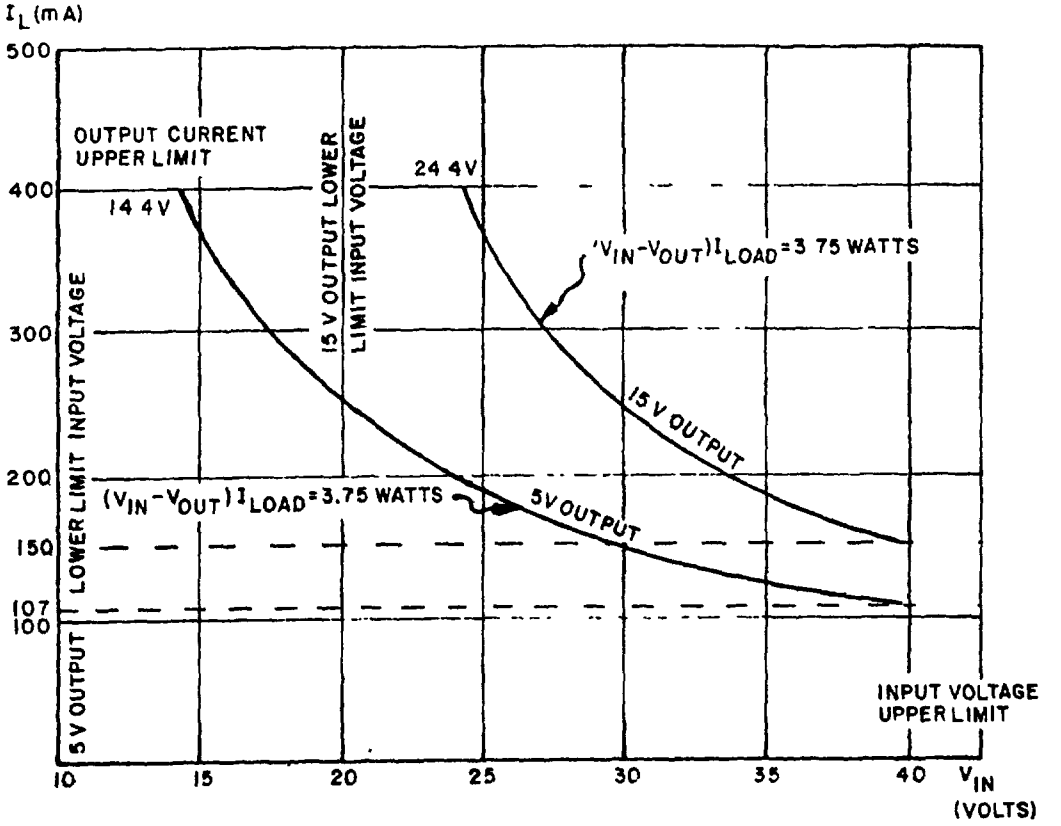


*FIGURE 2 Power dissipation limits.

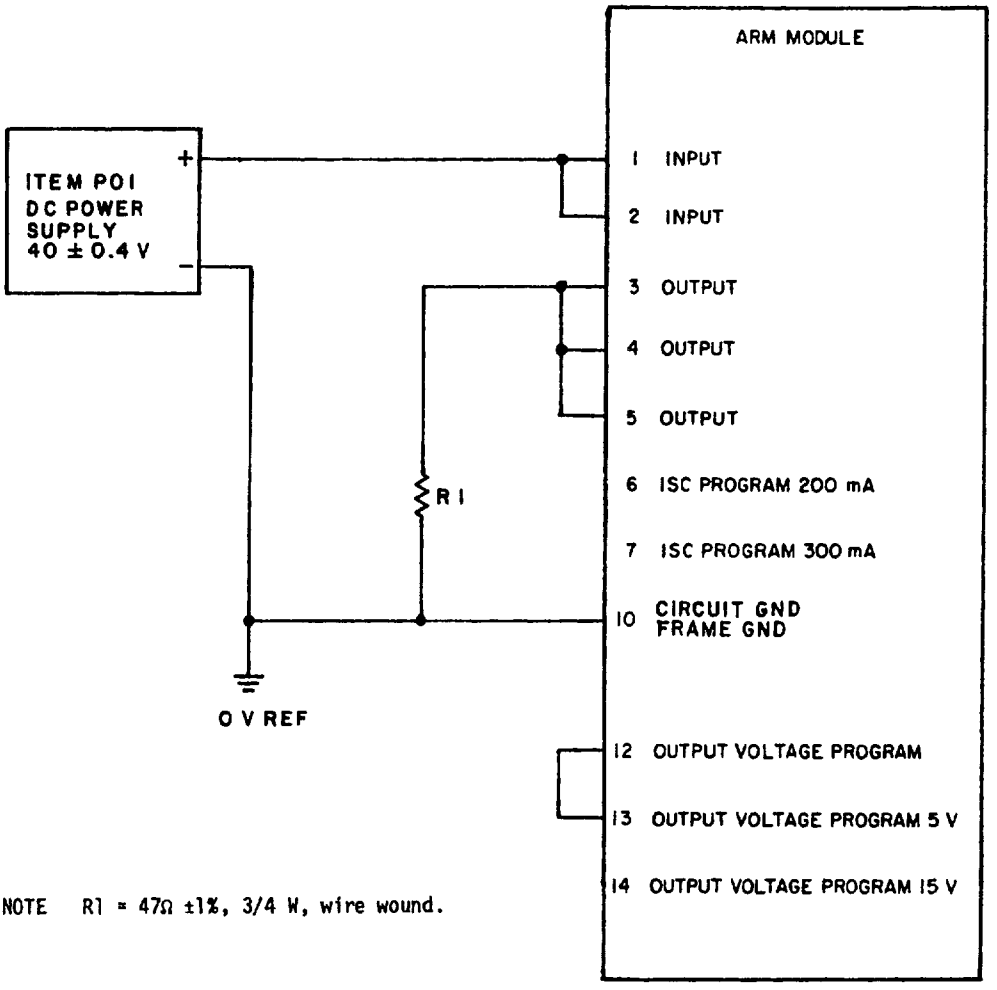
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE R1 = 47Ω ±1%, 3/4 W, wire wound.

*FIGURE 3. Operating temperature, component temperature, and life test circuit.

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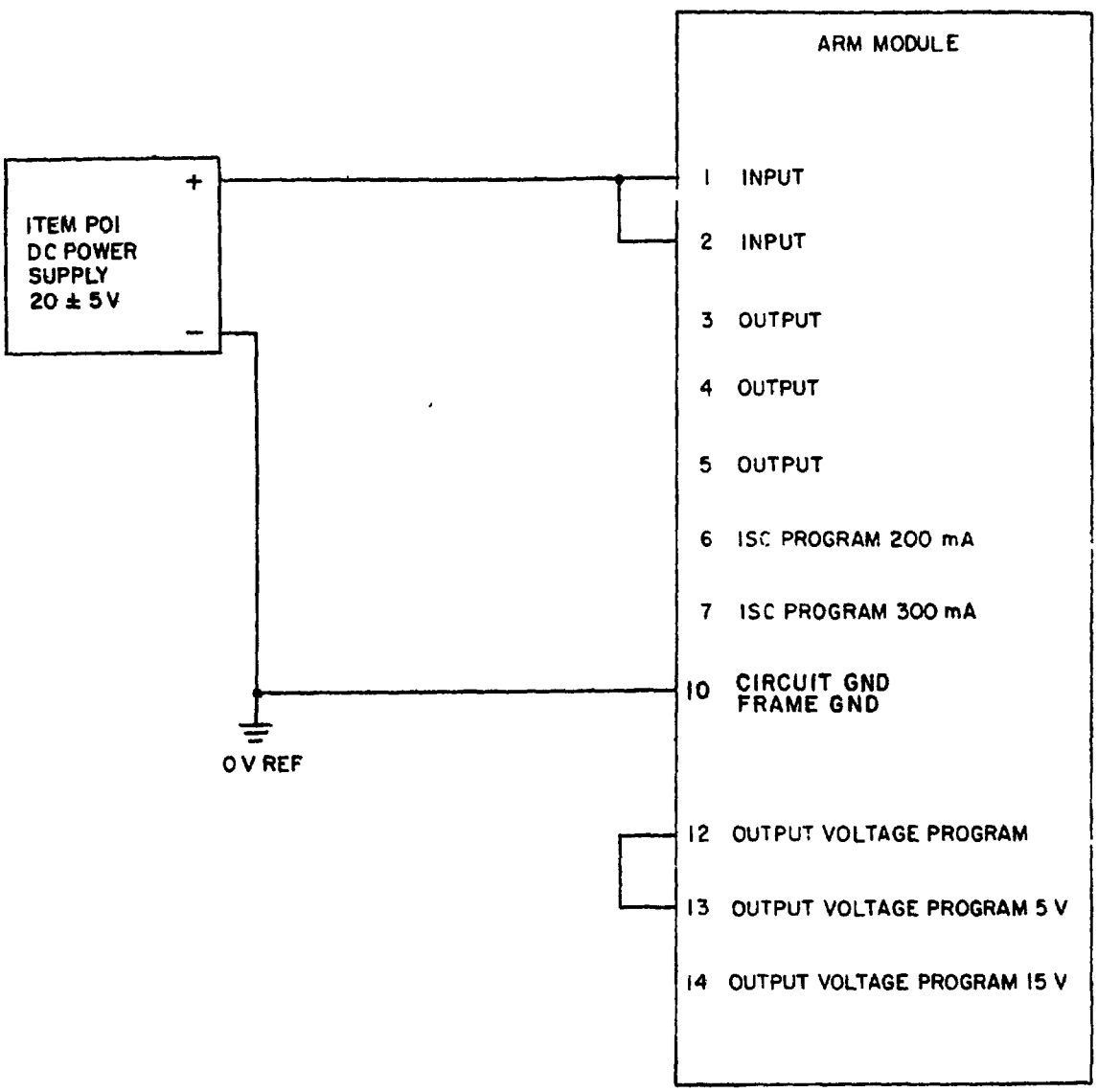


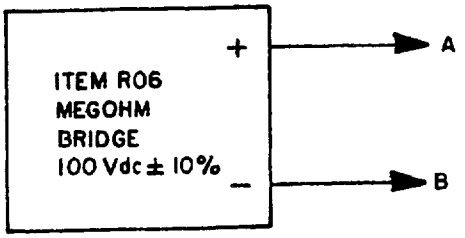
FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	1,2,3,4,5,6,7, 10,11,12,13,14	8,9,15,16 17,18,19,20	10	∞	MΩ	5
A2		8	9,15,16,17,18, 19,20				
A3		9	15,16,17,18, 19,20				
A4		15	16,17,18,19,20				
A5		16	17,18,19,20				
A6		17	18,19,20				
A7		18	19, 20				
A8	Isolation	19	20	10	∞	MΩ	5

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

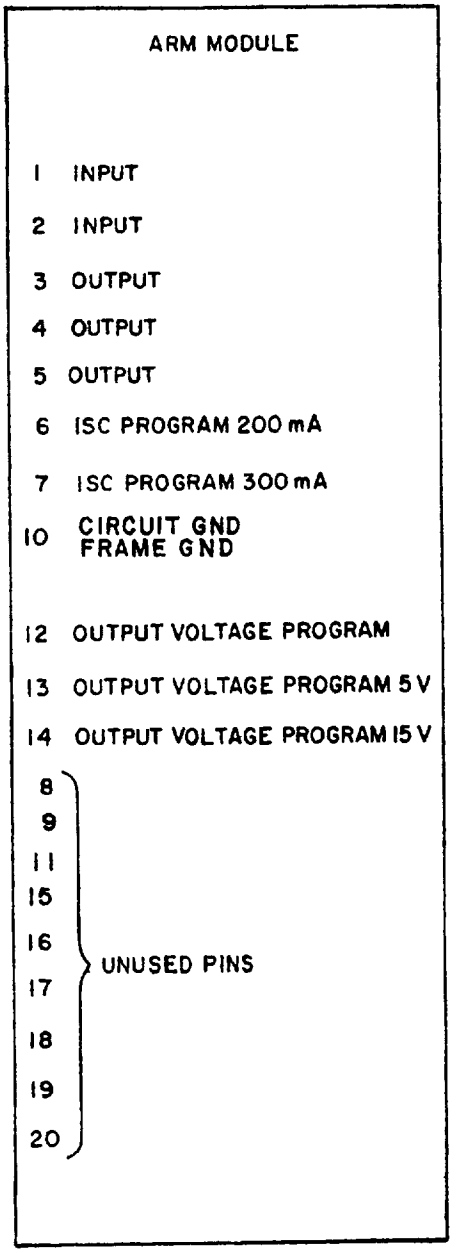


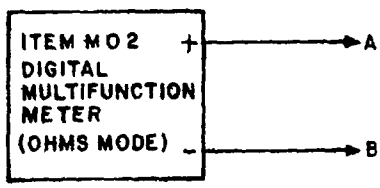
FIGURE 5. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity	1	2	0.0	1.0	Ω	6
B2	↓	3	4	↓	↓	↓	↓
B3		3	5				
B4		4	5				
B5	Continuity	10	Frame	0.0	1.0	Ω	6

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ARM MODULE

1	INPUT
2	INPUT
3	OUTPUT
4	OUTPUT
5	OUTPUT
6	ISC PROGRAM 200 mA
7	ISC PROGRAM 300 mA
10	CIRCUIT GND FRAME GND
12	OUTPUT VOLTAGE PROGRAM
13	OUTPUT VOLTAGE PROGRAM 5 V
14	OUTPUT VOLTAGE PROGRAM 15 V

NOTE: For test condition B5 use a sharp needle-like probe, pierce (do not scratch) the insulated surface coating of the module fin and measure the resistance between pin 10 and the module fin

FIGURE 6. Continuity test circuit for table III

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TABLE IV. Output voltage program test.

Test condition	Parameter	P01 (Vdc)	S1 position	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
C1	V_{OUT}	10 ±0.1	1	6	13	4.95	5.05	V	7
C2	V_{OUT}	20 ±0.2	2	6	14	14.70	15.30	V	7

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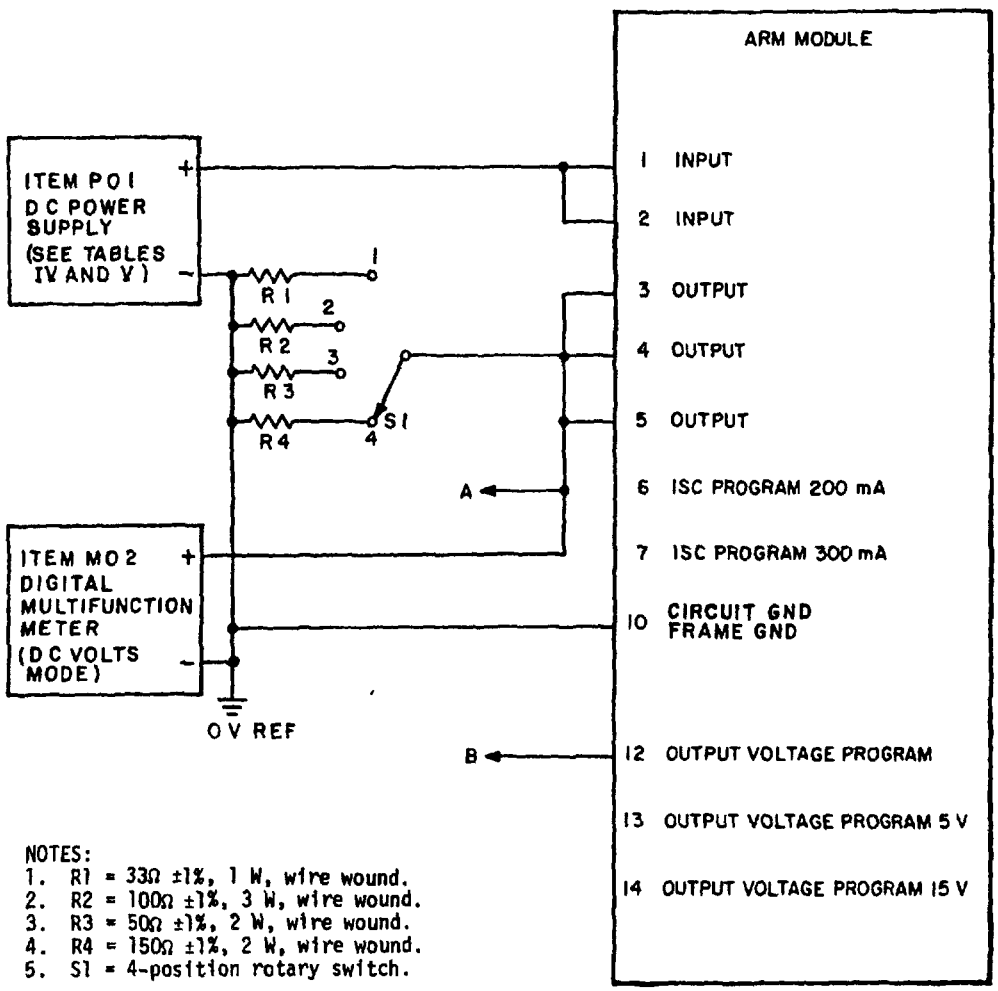
* TABLE V. DC input voltage regulation test.

Test condition	Parameter	POI (Vdc)	SI position	Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
D1	V_{OUT}	10 ±0.1	3	6	13	4.95	5.05	V	7
D2		15 ±0.15				← 1/ →			
D3		20 ±0.2							
D4		25 ±0.25							
D5		30 ±0.3							
D6		35 ±0.35							
D7		40 ±0.4	3		13	← 1/ →			
D8		20 ±0.2	4		14	14.70	15.30	V	
D9		25 ±0.25				← 2/ →			
D10		30 ±0.3							
D11		35 ±0.35							
D12	V_{OUT}	40 ±0.4	4	6	14	← 2/ →			7

1/ V_{OUT} should not vary more than ±15 mV from value of D1.

2/ V_{OUT} should not vary more than ±35 mV from value of D8.

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NOTES:
1. R1 = 33Ω ±1%, 1 W, wire wound.
2. R2 = 100Ω ±1%, 3 W, wire wound.
3. R3 = 50Ω ±1%, 2 W, wire wound.
4. R4 = 150Ω ±1%, 2 W, wire wound.
5. S1 = 4-position rotary switch.

*FIGURE 7 Output voltage program and dc input voltage regulation test circuit for tables IV and V.

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TABLE VI. Output short circuit current test

Test condition	Parameter	P01 (Vdc)	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
E1	Output short circuit current	10 ±0.1	Open	13	96.3	117 7	mAdc	8
E2	↓	20 ±0.2	Open	14	96.3	117.7	↓	↓
E3		10 ±0.1	6	13	183.6	224 4		
E4		20 ±0.2	6	14	183 6	224.4		
E5		10 ±0.1	7	13	276.4	337.6		
E6		20 ±0.2	7	14	276.4	337.6		
E7	↓	10 ±0 1	6, 7	13	363.6	444.4	↓	↓
E8	Output short circuit current	20 ±0.2	6, 7	14	363.6	444.4	mAdc	8

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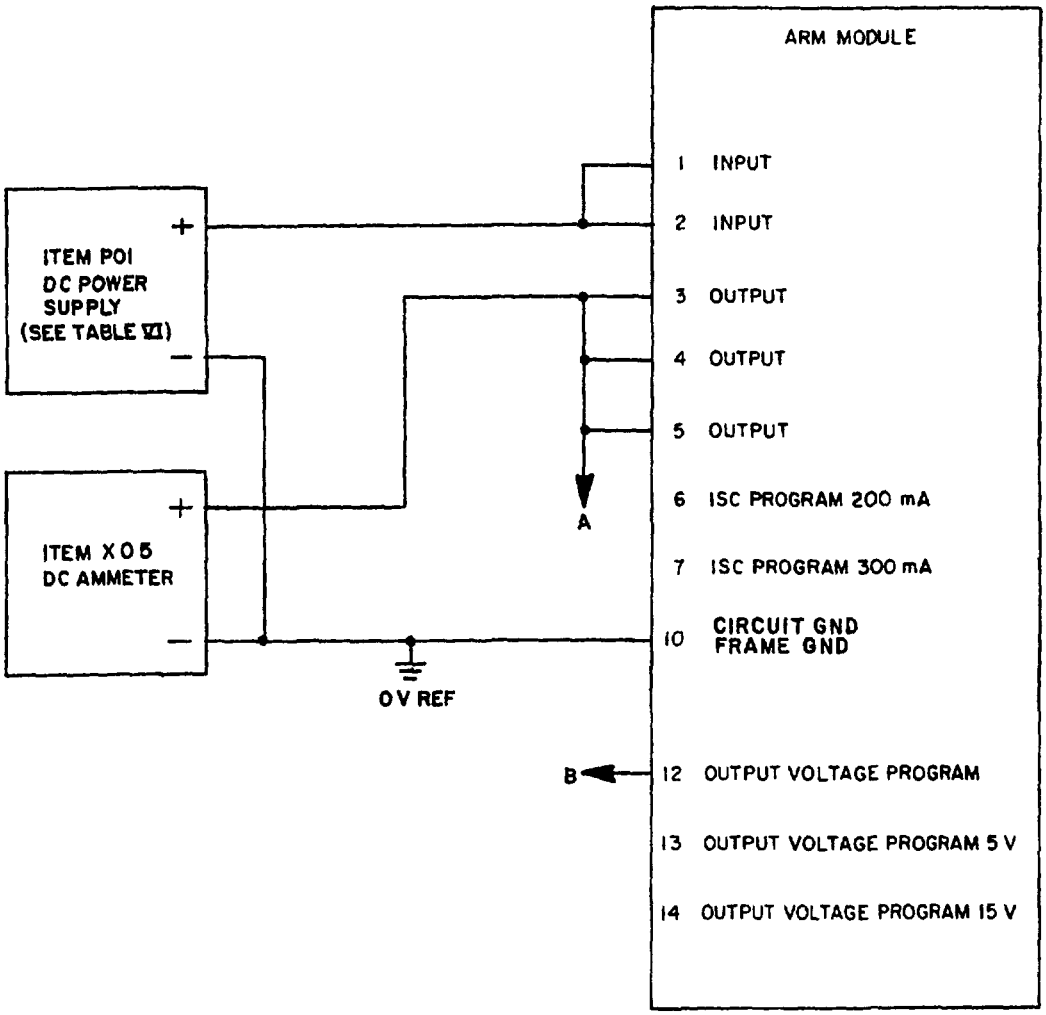


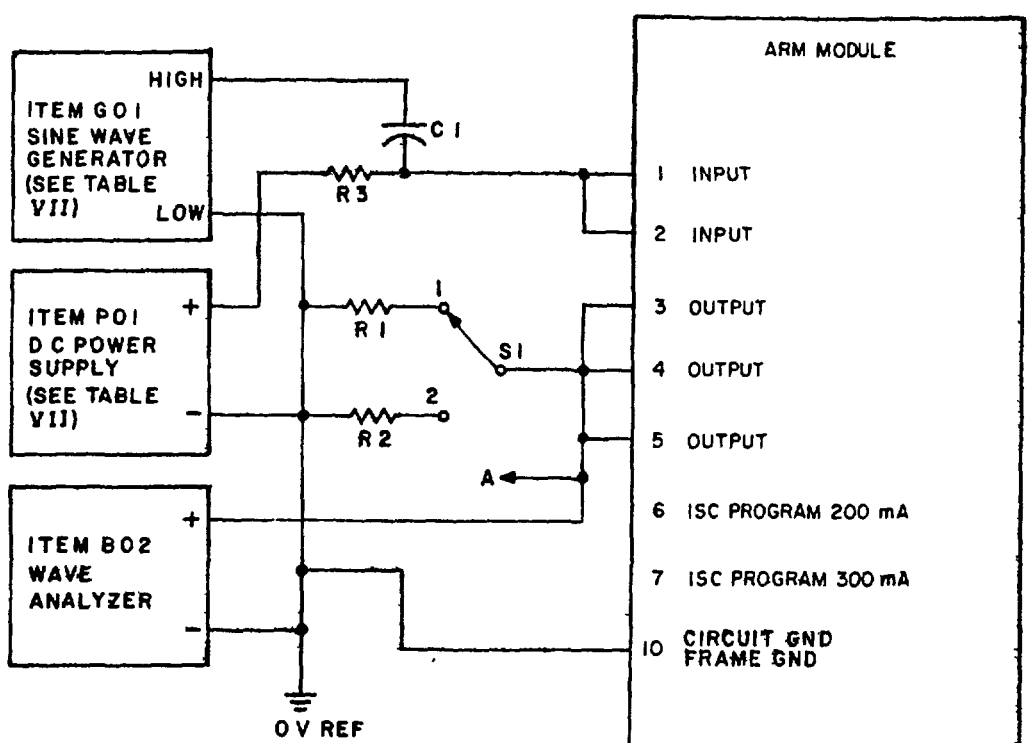
FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Input modulation regulation test

Test condition	Parameter	G01 (Hz)	P01 (Vdc)	S1 position	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
					A to pin	B to pin	Min	Max		
F1	V_{OUT}	120 ±12	Note 6	1	6	13	0.1	750	µVrms	9
F2	↓	800 ±80	Note 6	1	↓	13	0.1	750	µVrms	↓
F3	↓	120 ±12	Note 7	2	↓	14	0.01	2.25	mVrms	↓
F4	V_{OUT}	800 ±80	Note 7	2	6	14	0.01	2.25	mVrms	9

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NOTES:
1. R1 = 50Ω ±1%, 5 W, wire wound.
2. R2 = 150Ω ±1%, 2 W, wire wound.
3. R3 = 47Ω ±1%, 3/4 W, wire wound
4. C1 = 22 µF ±20%, 35 V, electrolytic.
5. S1 = SPDT.
6. Adjust item P01 to give 12 ±0.12 Vdc at pin 1 with pin 1 connected.
   Adjust item G01 to give 1.0 ±0.1 Vrms at pin 1 with pin 1 connected.
7. Adjust item P01 to give 24 ±0.24 Vdc at pin 1 with pin 1 connected.
   Adjust item G01 to give 1.0 ±0.1 Vrms at pin 1 with pin 1 connected.

*FIGURE 9. Input modulation regulation test circuit for table VII.

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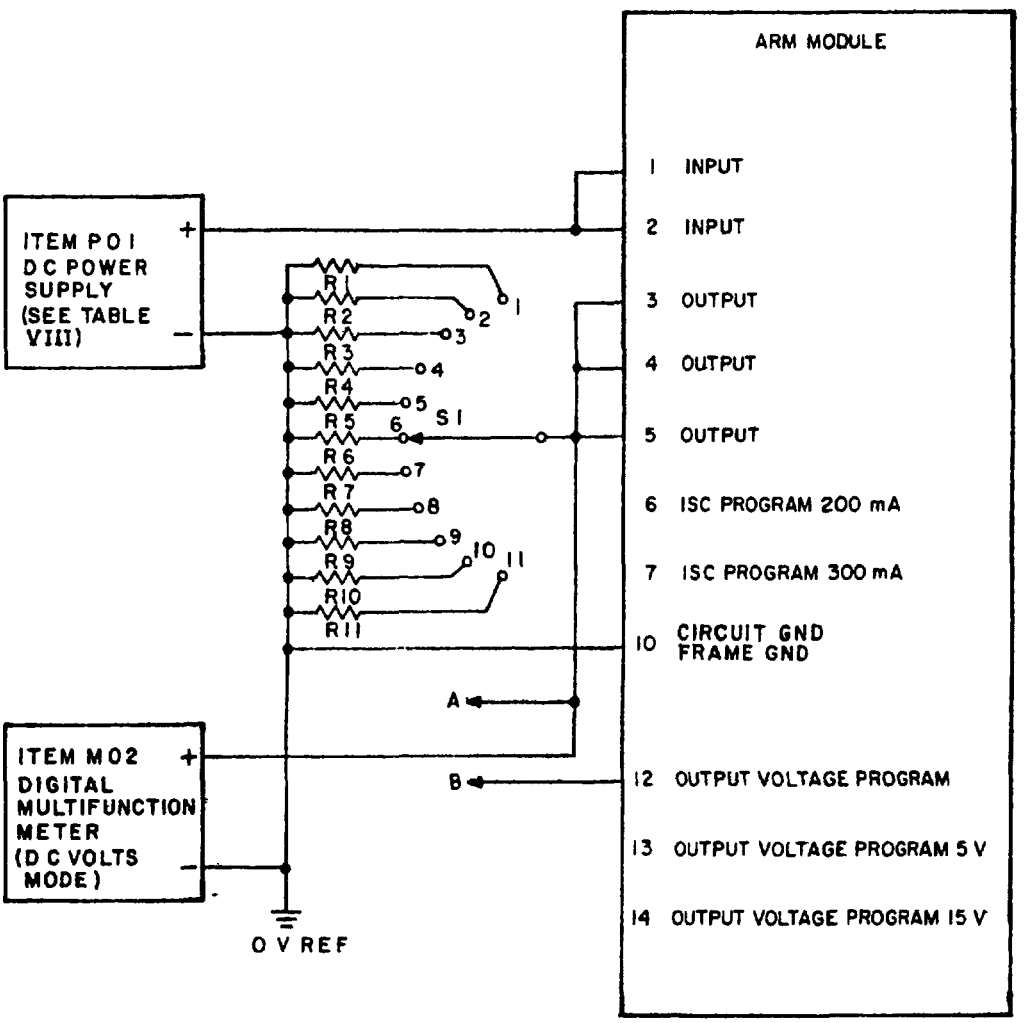
* TABLE VIII. Output load regulation test.

Test condition	Parameter	POI (Vdc)	S1 position	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
G1	V_{OUT}	10 ±0.1	1	Open	13	4.95	5.05	V	10
G2			2	Open		← 1/ →			
G3			3	Open					
G4			4	6					
G5			5	6					
G6			6	7					
G7		10 ±0.1	7	6, 7	13	← 1/ →			
G8		20 ±0.2	8	Open	14	14.70	15.30	V	
G9			9	Open		← 2/ →			
G10			10	Open					
G11			11	6					
G12			2	6					
G13			3	7					
G14	V_{OUT}	20 ±0.2	4	6, 7	14	← 2/ →			10

1/ V_{OUT} should not vary more than ±10 mV from Value of G1.

2/ V_{OUT} should not vary more than ±30 mV from Value of G8.

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NOTES:
1. R1 = 4.99 kΩ ±1%, 1/10 W.
2. R2 = 100Ω ±1%, 4 W, wire wound.
3. R3 = 68Ω ±1%, 4 W, wire wound.
4. R4 = 50Ω ±1%, 5 W , wire wound.
5. R5 = 33Ω ±1%, 1 W, wire wound.
6. R6 = 22Ω ±1%, 2 W, wire wound.
7. R7 = 17Ω ±1%, 2 W, wire wound (use 22Ω ±1%, 2 W, in parallel with 75Ω ±1%, 1/2 W).
8. R8 = 15 kΩ ±1%, 1/10 W.
9. R9 = 300Ω ±1%, 1 W, wire wound.
10. R10 = 200Ω ±1%, 2 W, wire wound.
11. R11 = 150Ω ±1%, 2 W, wire wound.
12. S1 = 11-position rotary switch.

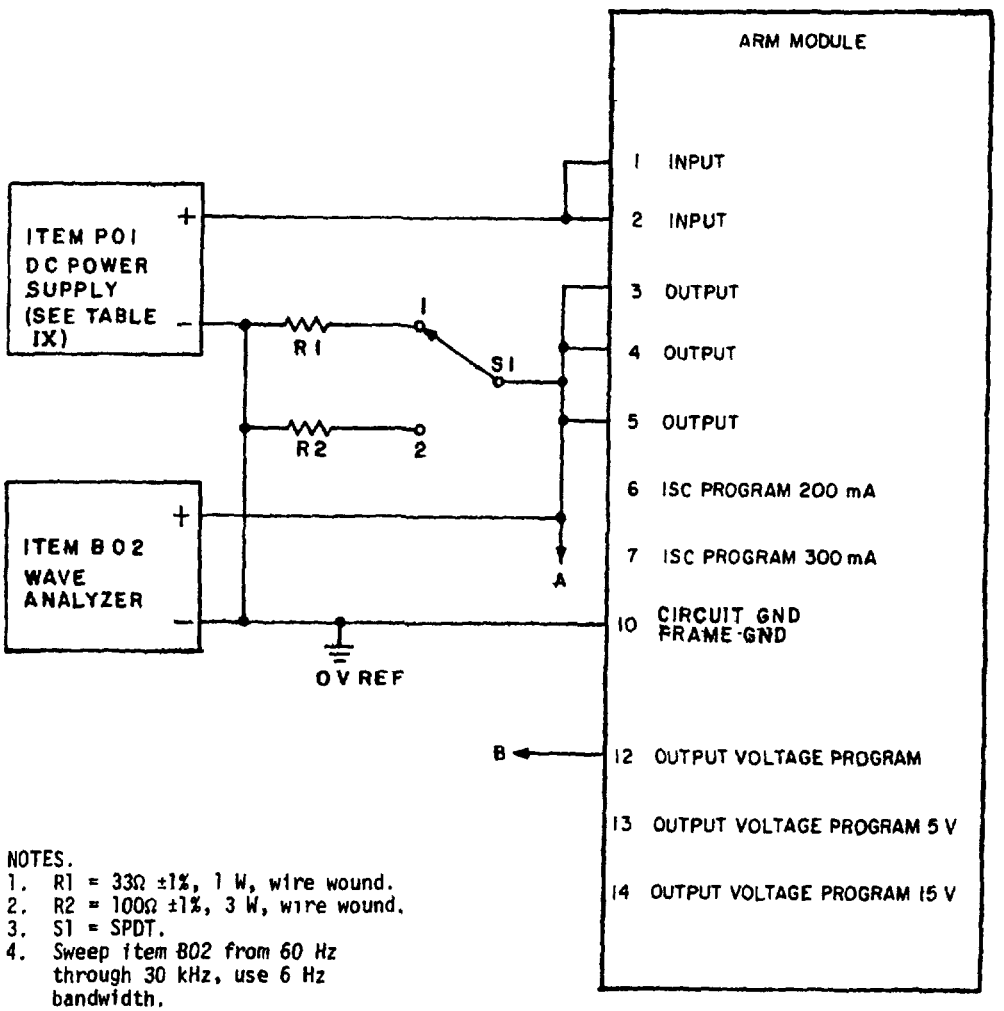
*FIGURE 10 Output load regulation test circuit for table VIII.

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TABLE IX. Output noise test.

Test condition	Parameter	P01 (Vdc)	S1 position	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	Min	Max		
H1	Output noise	10 ±0.1	1	6	13	0.0	250.0	µVrms	11
H2	Output noise	20 ±0.2	2	6	14	0.0	250.0	µVrms	11

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NOTES.
1. R1 = 33Ω ±1%, 1 W, wire wound.
2. R2 = 100Ω ±1%, 3 W, wire wound.
3. S1 = SPDT.
4. Sweep item B02 from 60 Hz through 30 kHz, use 6 Hz bandwidth.

*FIGURE 11. Output noise test circuit for table IX.

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* TABLE X. Output impedance check 1/ - 3/.

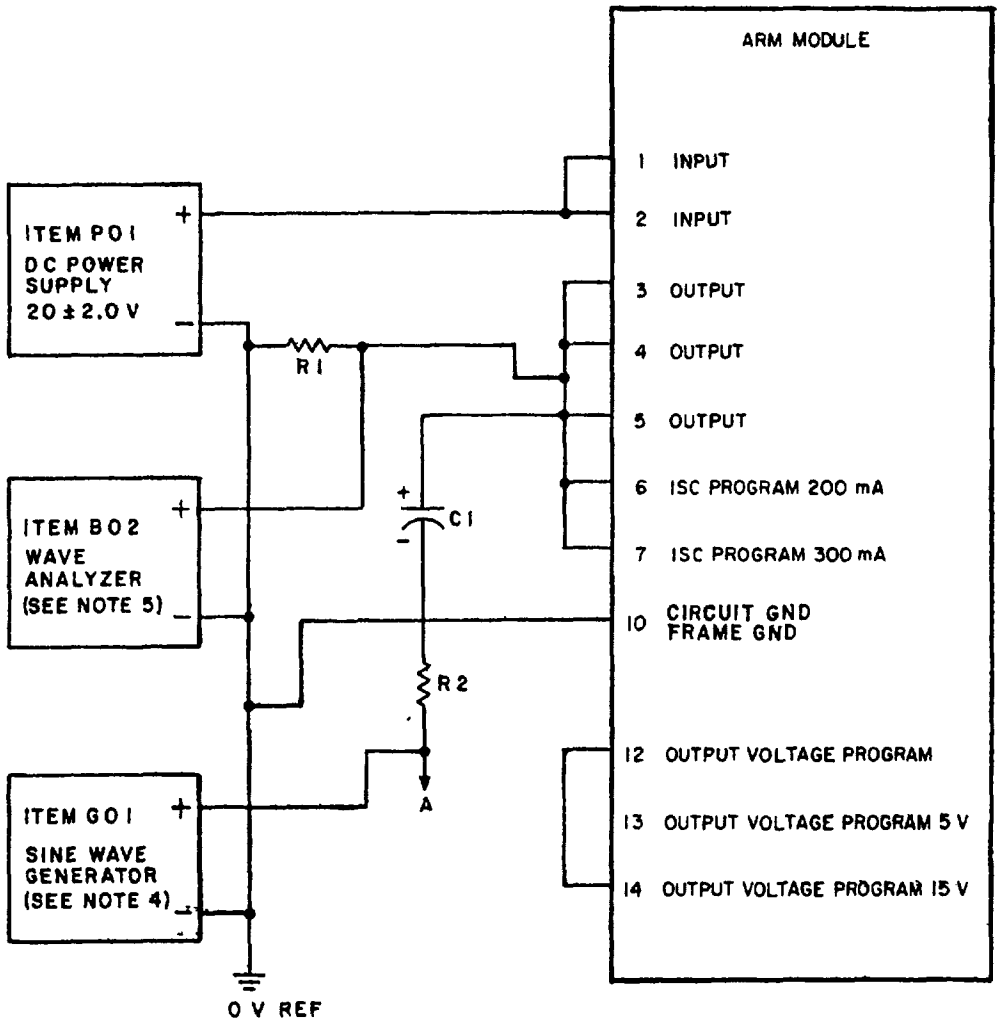
Test condition	Parameter	GOI	Initial limits 25° ±5°C Ambient		Unit	Test figure
			Min	Max		
J1	Output impedance	100 Hz ±1%	5 (1)	500 (100)	µVrms MΩ	12
J2	Output impedance	1 kHz ±1%	5 (1)	500 (100)	µVrms MΩ	12
J3	Output impedance	50 kHz ±1%	10 (2)	1,000 (200)	µVrms MΩ	12

1/ Output impedance = $\frac{V_{OUT}rms}{V_{IN}rms/R2}$

2/ V_{OUT} rms = pin 4.

3/ V_{IN}rms = point A.

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NOTES.
1. R1 = 50Ω ±1%, 5 W, wire wound.
2. R2 = 10Ω ±1%, 1/4 W.
3. C1 = 1,000 µF ±20%, 25 V, electrolytic.
4. Adjust item G01 to give 50 ±5 mVrms at point A at the frequency specified in table X.
5. Adjust item B02 to item G01 frequency.

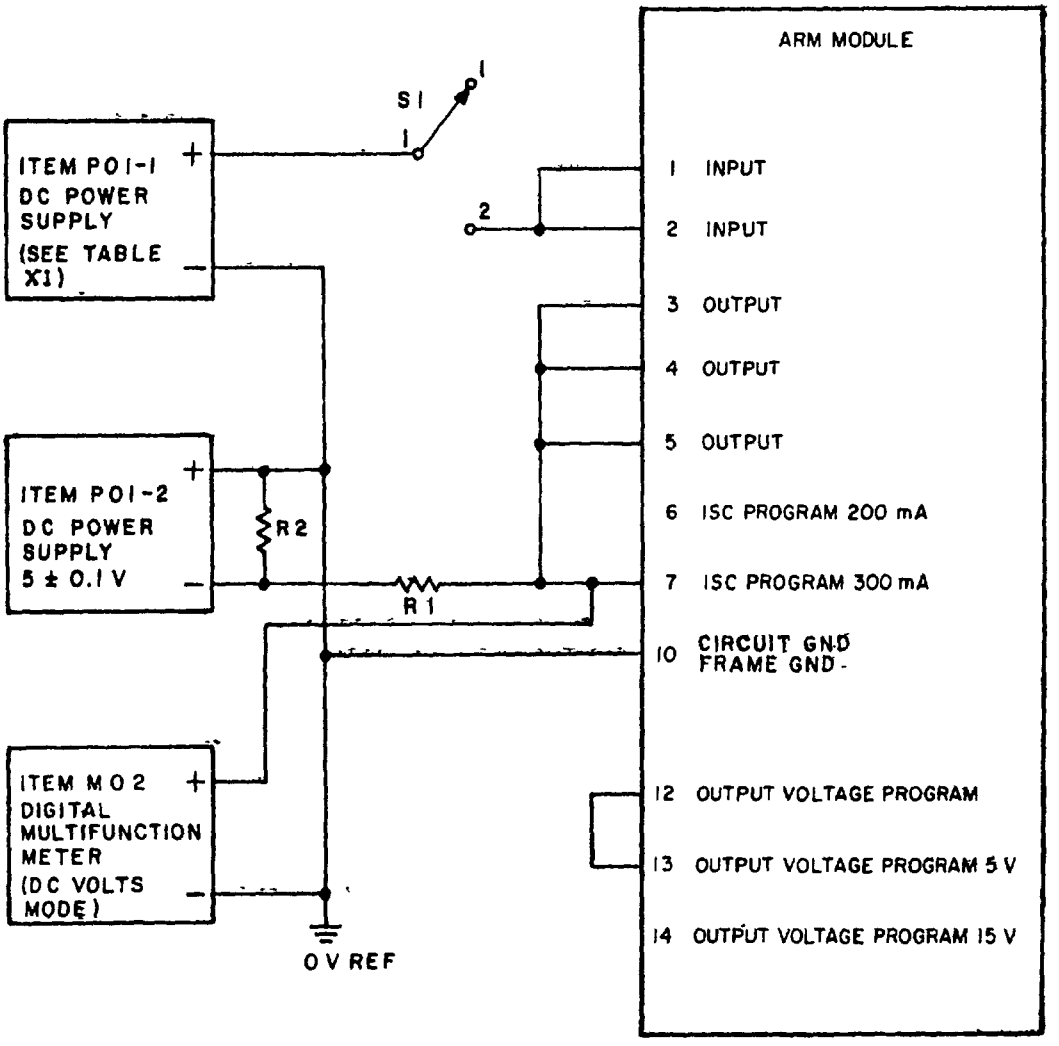
*FIGURE 12 Output impedance check test circuit for table X.

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TABLE XI. Output voltage latch test.

Test condition	Parameter	P01-1 (Vdc)	S1 position	Initial limits 25° ±5°C Ambient		Unit	Test figure
				Min	Max		
K1	V_{OUT}	10 ±1.0	1	-0.950	-0.650	V	13
K2	V_{OUT}	10 ±1.0	2	4.95	5.05	V	13

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NOTES:
1. R1 = 50Ω ±1%, 2 W, wire wound.
2. R2 = 50Ω ±5%, 5 W, wire wound
3 S1 = SPST.

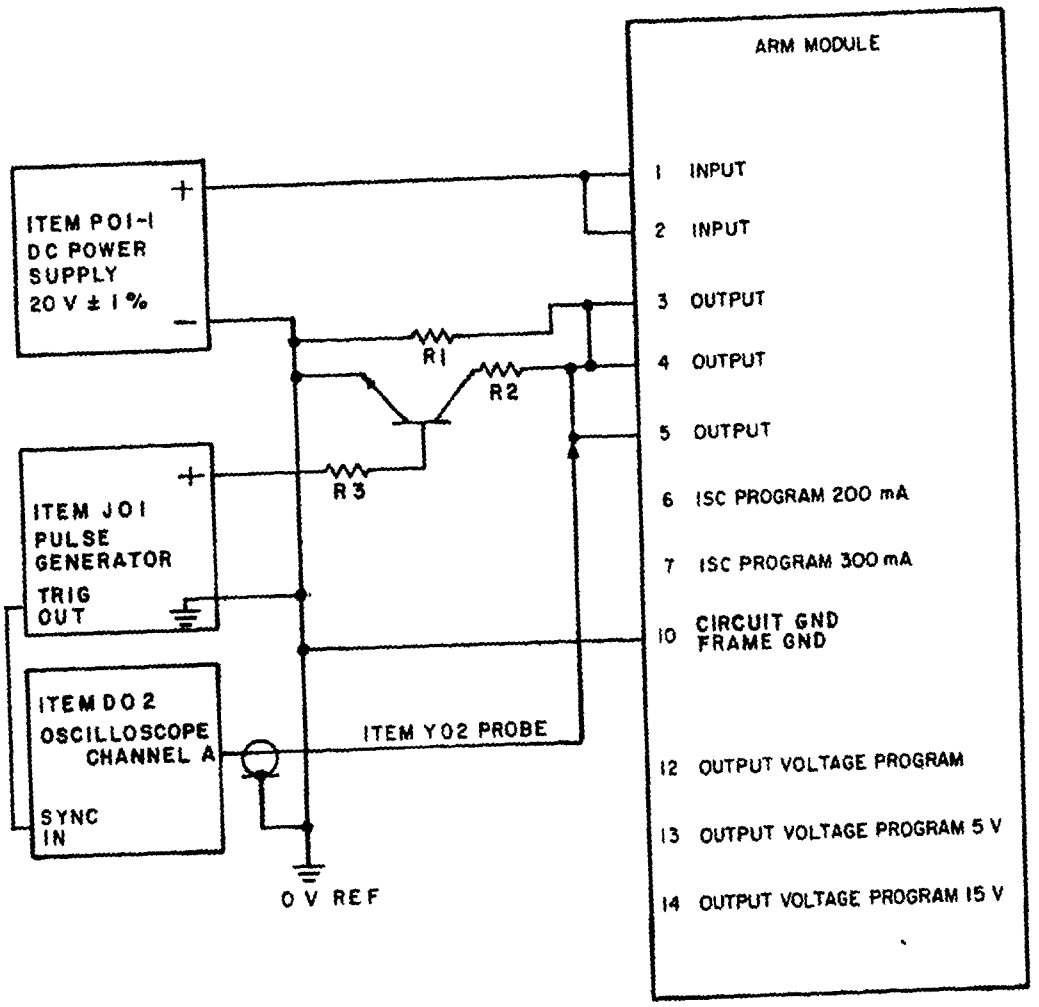
*FIGURE 13. Output voltage latch test circuit for table XI

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TABLE XII. Load transient response test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		Pin 12 to pin	Pin 7 to pin	Min	Max		
L1	Load transient response	13	Open	0	150	mV	14, 15
L2	Load transient response	14	5	0	450	mV	14, 15

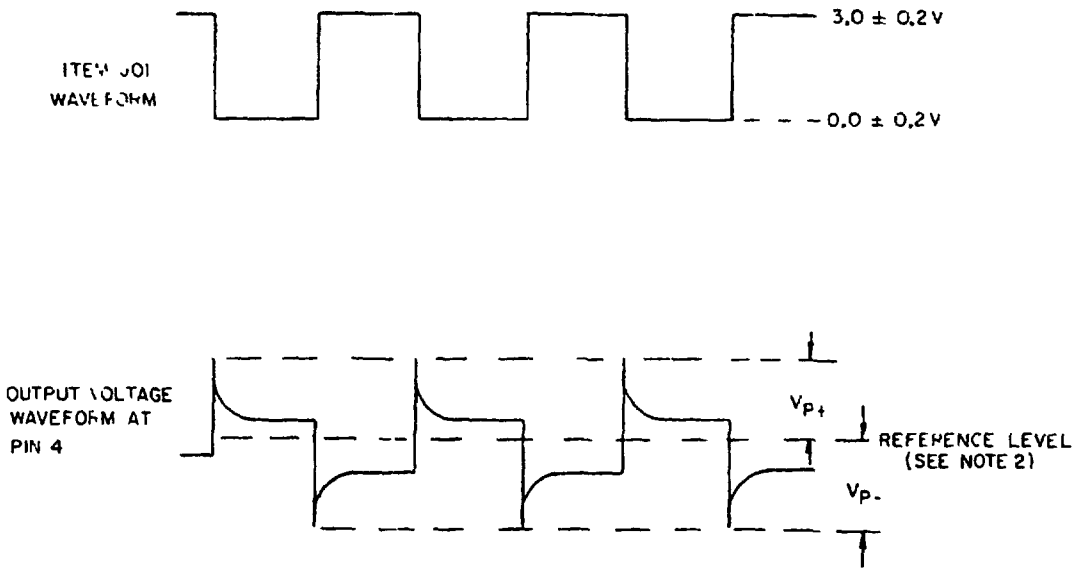
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NOTES:
1. R1 = 200Ω ±1%, 2 W, noninductive, wire wound.
2. R2 = 100Ω ±1%, 3 W, noninductive, wire wound
3. R3 = 169Ω ±1%, 1/8 W.
4. Q1 = 2N2364A, or equivalent.

*FIGURE 14. Load transient response test circuit for table XII.

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NOTES:
1. Item J01 setup:
   PW = 50 µs ±10%
   PRP = 100 µs ±10%
   $t_{THL} = t_{TLH} = 10 \pm 5$ ns
2. The reference level is the +5 V or +15 V level obtained before the item J01 pulse is applied to the circuit.
3. The data recorded for test conditions L1 and L2 is the largest of the two readings V_{p+} and V_{p-}.

FIGURE 15 Load transient response test circuit waveforms for table XII

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